

SANDEEP SAHDEV



GETTY IMAGES

**1. Field of gold**
A wheat farmer in Punjab; (inset) M.S. Swaminathan

**2. Milky way**
An Amul milk bottling factory near Ahmedabad; (inset) Verghese Kurien

Indian plant geneticist M.S. Swaminathan played a critical role not only in facilitating the import of these seeds but also making them suitable for the Indian climate. That was the first big milestone of Indian agriculture. Both these stellar scientists, Borlaug and Swaminathan, have been globally recognised for their contribution to saving millions of lives through their scientific contribution.

But how many of us are familiar with the names of US plant breeder Henry Beachell and Indian agri-scientist Gurdev Khush, both—working at the International Rice Research Institute in Los Baños, Philippines—who brought about similar revolutionary changes in rice production by developing dwarf varieties of **high-yielding rice**? Both got the World Food Prize for their contributions to the development of HYVs of rice. This would be the second biggest achievement because of its impact on India. It was because of these breakthroughs in wheat and rice during the second half of the 1960s through the mid-mid-'80s that India experienced a Green Revolution, giving it much-needed self-sufficiency in basic staples. In 2021-22, India exported about 31mmt of cereals (21mmt of rice, 7mmt of wheat, and about 3mmt of maize). India's exports of rice constituted almost 40 per cent of global rice exports, making it the biggest player in that arena.

The third and even bigger revolution took place in India's milk production. It was initially steered by Verghese Kurien under **Operation Flood**, starting in the mid-'70s. And three decades later, after the liberalisation of milk processing by the Vajpayee government in 2002-03,